

Dynamic Re-Mining: Design and Test Plan

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1 The problem

We mine factors once, from a direction someone types in, and then trade them forever. That doesn't hold up.

2 What already works: refitting the combiner

Refit the combiner every year on data before the year being traded. Same factors throughout, nothing re-mined, so any gain comes from the refit alone.

Table 1: MAIN-FULL, 147 factors. Fit 2005–2016, scored 2017–2025; LightGBM combiner, top-50 book, full cost model. Every return is excess over the cap-weighted CSI300 total-return index, so CRR is cumulative excess return. Frozen is the single 2016 fit; refit is a fresh fit each year on data before the year being traded, with the factor library unchanged throughout.

year	frozen 2016 fit		annual refit		Δ ARR
	IC _{rank}	ARR	IC _{rank}	ARR	
2021	0.0504	+5.82%	0.0586	+11.40%	+5.58 pp
2022	0.0524	+1.81%	0.0598	+8.98%	+7.17 pp
2023	0.0160	−7.42%	0.0298	−1.99%	+5.43 pp
2024	0.0288	−4.31%	0.0533	−3.75%	+0.56 pp
2025	0.0469	−14.94%	0.0616	−10.25%	+4.69 pp
mean	0.0389		0.0526		+4.69 pp
CRR		−18.8%		+2.68%	+21.5 pp

All five years improve, on both rank IC and return. Mean rank IC rises 35%. This is the only configuration out of the nine we have tested with a positive CRR, and we don't currently do it — production fits once and never refits again.

2023 is worth a look. A refit more than halves the loss and lifts rank IC by 86%. About half of what reads as a regime break is just a stale combiner.

Refreshing the *factors* is worth roughly twice as much again: fresh factors score 0.1001 and 0.1136 against 0.0526 for stale factors with a fresh combiner. Putting both together, $IR = TC \cdot IC \cdot \sqrt{N}$ goes 0.30 (both stale) $\rightarrow$ 0.36 (refit only) $\rightarrow$ 0.68 (re-mined and refit), which is the first number above the usual 0.5 bar. There is a caveat on that figure in §4.1.

3 The design

Two layers. A deterministic one that always runs, and an agent that can only make it act sooner.

Deterministic. Each factor's rolling 63-day IC is compared to its validation IC at admission. Healthy at 70% of baseline or better. Soft decay below 70% for 30 days straight, which halves its weight. Hard decay below 50% for 60 days, which drops it from scoring but keeps the UUID and re-tests it quarterly. Monitor daily, refit monthly, re-mine at least annually and sooner if a third of the book hard-decays. Annual is the floor, not a safe choice: the published half-life for this kind of factor is now around 18 months.

The agent. The deterministic rule can’t fire until a breach has held for 30 days, so there’s a month or two of lag to claw back. The agent runs weekly and on every tier change. It can pull a refit, a re-mine, or a demotion forward. The only thing it can delay is a soft demotion, and only with a computed market condition attached, expiring after a quarter, capped at a quarter of the book, and never against hard decay. If it breaks or returns nonsense, nothing happens — the deterministic layer carries on.

Soft decay under a tree model. Halving a feature’s weight does nothing to a gradient-boosted tree, and LightGBM gives us no per-factor weights anyway. The combiner already fits one model per seed and averages them, so a soft-decayed factor goes into half the seed models instead of all of them. Same effect, no new machinery.

The library. Keep everything, forever, admitted or rejected, under its UUID. What actually trades is a capped subset chosen at each refit: drop the hard-decayed, drop the broken, then enforce an orthogonality gate so no two bets are the same, then cap and replace. The orthogonality gate exists in the code and is switched off by default.

Directions. Each re-mine writes its own, from statistics computed up to that date: dispersion, breadth, the size-decile spread, volatility, return autocorrelation, turnover, overnight gaps, industry dispersion, correlation. No news. The 2023 rotation *is* rising dispersion and an inverted size spread, and the agent sees those as numbers with no label on them. We never tell it the story.

Parents. Today selection is pure exploitation, from a pool thrown away at the end of each run; it has produced operator monoculture twice, and crossover measures worse than mutation (median absolute rank IC 0.0067 against 0.0114). The library becomes an archive binned by operator class, lookback horizon, regime profile and decay tier, with parents drawn on a fixed, logged budget: 50% the best performers by shrunk marginal contribution, 30% occupants of thinly populated bins whatever their fitness — that is what breaks the monoculture — and 20% dead factors whose regime matches the current state. Within a pool we pick cells by UCB, not factors: verdicts are noisy enough that three of four flip across combiner seeds, and a cell pools many trials.

4 How we test it

Replay history. At each trigger point everything uses data up to that date only, the next year is scored and never looked at again; returns stitch into one series per arm.

arm	directions	schedule	what it tells us
C	—	none	static baseline, CRR -41.3%
D	—	calendar	refit only, CRR $+2.68\%$
A	fixed	deterministic	whether re-mining is worth it
B	agent	deterministic	the direction agent
G	fixed	+ decay agent	the decay agent
H	agent	+ decay agent	both together
H [−]	agent	+ agent, no exemptions	what the regime exemption buys
E	Alpha158	calendar	leakage control — public handcrafted set, no LLM
F	random DSL	calendar	same, without Alpha158’s survivorship

A, B, G and H are a 2×2 over directions and scheduling: $A - D$ is re-mining, $B - A$ the direction agent, $G - A$ the decay agent, $(H - G) - (B - A)$ whether the two agents help or hinder each other. C and D are already measured; E and F need no mining.

Cost is measured, not guessed. `meanvar_20260828_194432`, the 150-factor library this document is built on, took **15.1 hours of compute** (160 batches at 5.7 minutes); its 51 hours of wall clock included a 34-hour interruption, so a replay is budgeted on machine time. At that rate the scheduled

floor of 18 mines is 11 days, the realistic full-system test with both agents live is **22–26 mines, 14–16 days**, H^- adds about five more if H beats G, and the reduced three-point set is 6 mines, under four days.

Eighteen counts only the scheduled re-mines, so it is a floor, not the estimate. The nine nominal trigger points are the first trading day of 2017 through 2025, but that is the ceiling on how long the deterministic layer waits, not the dates it mines on: it also fires when a third of the active set has hard-decayed or the book’s 63-day rank IC has held below half its baseline for 60 days, and the decay agent pulls those forward but never pushes them back. The sample holds four down years and the 2023 trough, so they do fire — which is where 22–26 comes from, A and B mining at least nine times each and G and H at least as often, a mine being reused wherever two arms land on the same date.

The reduced set is the start of 2016, 2019 and 2022, blocks of three, three and four years; it begins in 2016, so arm C would need re-cutting over 2016–2025 to match.

Headline metric is stitched CRR. Alongside it, per year and per arm: ARR, IR, MDD, rank IC, IC, ICIR, TC, turnover, cost, book size, admissions, tier counts. The benchmark’s own return sits next to every arm, because excess return here is driven by beta and an arm that wins in a falling market hasn’t necessarily found anything.

4.1 Leakage

The model’s weights contain the future and no prompt takes that back. We bound what it can do — only the training window is disclosed, a validator strips dates and tickers, and a direction names a mechanism rather than an answer — and then we measure two things.

Late-sample leakage is already ruled out. If hindsight were being used, the factors would hold up later in the sample. Instead rank IC falls steadily and bottoms out in 2023, the most heavily documented regime break we have and exactly where the model would know the most. Admission also selects on past data, so a factor that only works in the future never gets in.

The first year after a fit is not ruled out, and it matters more. Among factors that work before the cutoff, hindsight can still pick the ones it knows last longest. That inflates year one and fades as those picks decay, which looks exactly like what we see. Arms A and B can’t detect it either, since both write factors with the same model and the effect cancels. And the whole case for re-mining rests on a first-year number.

Hence arms E and F, which have no LLM in them at all: E is Alpha158, a public set of handcrafted factors, F is randomly composed expressions, and neither stands in for what a firm running this loop would trade — they show only what the pipeline produces with no model in it. Compare year-one IC against steady-state IC as a ratio. If ours is much higher than theirs, that’s the leak. Until then the number to expect is the steady state — 0.0526 mean Rank IC over 2021–2025 with a refreshed combiner, above the 0.03–0.05 band a usable signal usually occupies. It is the first-year 0.11 we don’t get to quote.

5 What this settles, and what it doesn’t

	2017–2025	2017	usable range
IC	0.0425	0.0941	0.02–0.05
Rank IC	0.0565	0.1104	0.03–0.05
ICIR	0.315	0.743	0.3–0.5
Rank ICIR	0.409	0.845	0.3–0.5

On clean out-of-sample data the library sits inside the normal range for a usable equity signal on every predictive-power metric, and its first year is above it. The system finds alpha. That was the research question and the answer is yes.

Where it goes wrong is construction. Take $IR = TC \cdot IC \cdot \sqrt{N}$: with IC 0.0425, $\sqrt{300} = 17.3$ and TC 0.37, gross IR comes out around 0.3. Each term on its own is ordinary. A TC of 0.37 is normal for a long-only book, which usually runs 0.3–0.6, so there is no hidden loss to go and recover. The product is too small because of the mandate — 300 names, 50 holdings, long only, measured against a cap-weighted index we have no way to hedge.

The clearest way to see it is to put our excess return next to the index’s, year by year, on the same annual-refit numbers as Table 1.

	2021	2022	2023	2024	2025
index	−3.5	−19.8	−9.1	+18.2	+21.0
our ARR, excess	+11.40	+8.98	−1.99	−3.75	−10.25
our ARR, absolute	+7.9	−10.8	−11.1	+14.5	+10.8

The absolute row is approximate: the eval path stores only the excess, so it is excess plus the index’s own return to first order. It makes the same point from the other side — the book moves with the market and keeps part of it, +14.5% and +10.8% in the two up years against the index’s +18.2% and +21.0%.

The excess is the row to read carefully: positive in 2021 and 2022, both years the index fell, and negative otherwise — including 2023, which fell 9.1% and still cost us 2.0 points. A falling index is not sufficient, but it is necessary, which is not something you would say about a book earning its return from stock selection.

Five things say the same:

1. Annual excess return correlates -0.73 with the benchmark’s own return and only $+0.11$ with our rank IC. A book holding no mined factors shows the same thing at -0.64 , so it’s the construction, not the factors.
2. Raising the position cap from 3% to 10% left TC at 0.36, and going dollar-neutral lifted it only to 0.51. The cap isn’t what binds — long-only is — and both still lost money.
3. Three of our nine configurations do worse than holding no factors.
4. Our best signal year lost money. 2017, rank IC 0.1104, the strongest this system has produced out of sample: -5.45% , because the index rose 24.25% .

The objective we optimise is a function of IC. Nothing in the system optimises IR or CRR. The design in this document fixes the research half properly — it watches its own decay, re-mines against it, refits continuously — and on the numbers above that should take gross IR from 0.30 to somewhere near 0.68.

It won’t fix a transfer coefficient that is already normal, or a benchmark we can’t track long-only, or a construction stage with no search, no admission rule and no multiple-testing control. Turning predictive power into profit needs a second system built for construction the way this one was built for factors, searching over constraints, hedges, weights and rebalancing, with realised net IR as the objective instead of IC.

If we’re wrong, the replay will say so: an arm that lifts IC should then lift CRR. Our measured correlation between annual excess return and rank IC is $+0.11$, so we don’t expect it to.

Factor side	Portfolio side
LLM search with mutation, crossover, refinement	<code>topk</code> = 50
Admission gate tested against its own noise	<code>max_weight</code> = 0.03
FDR control and deflated Sharpe	<code>n_drop</code> = 5
Orthogonality gate, diversity scoring, lineage	
Per-factor tear sheets and diagnosis	
A research system.	Three constants from one sweep on one 2021 window.

Not covered here: shorting and borrow costs, live news, real capital, any change to construction itself, and NAV, which is simulated and compounds from 100M CNY where the impact term is still small enough not to matter.